GLOBAL STRATEGY

ENERGY. PRODUCTIVE CAPACITY.
FINANCIAL RESILIENCE.

THE GLOBAL ECONOMIC INFRASTRUCTURE IMPERATIVE

A strategic briefing on the changing relationship between energy systems, physical infrastructure and capital.

RHEO

Making productive capacity measurable for capital.

NOVEMBER 2026 | STRATEGIC BRIEFING

EXECUTIVE PERSPECTIVE

The physical economy is becoming more strategically important.

The global economy is entering a period in which electricity demand, digital infrastructure, industrial capacity and energy security are becoming more tightly interconnected. At the same time, infrastructure investment must compete for capital under conditions of geopolitical uncertainty, fiscal constraints and changing financial risks.

The central issue is not simply whether more infrastructure will be built. It is whether productive capacity can be identified, assessed, financed and governed with sufficient confidence to deliver durable economic value.

This creates a structural opportunity for better connections between physical assets and financial capital. Rheo's proposition is to develop an independent financial infrastructure layer that helps make productive capacity more measurable, verifiable and financeable.

3.6%	US$3.4T	950 TWh
Forecast average annual global electricity-demand growth, 2026–2030 [1]	Estimated global energy investment in 2026 [2]	Projected data-centre electricity use in 2030 [3]

These figures describe forecasts and estimates, not guaranteed outcomes. They indicate the scale and direction of structural change; they do not by themselves prove demand for any single company or solution.

The strategic thesis

Productive capacity must become easier to measure, verify, account for and finance. This is relevant across developed and emerging economies, although the specific needs, institutions and regulatory requirements differ by market.

01 / THE GLOBAL LANDSCAPE

Electricity is becoming a more central economic input.

The International Energy Agency (IEA) forecasts global electricity demand growth averaging 3.6% annually from 2026 to 2030. It expects the next five years to add electricity demand at a pace around 50% higher than the average of the previous decade. Industry, electric vehicles, cooling and data centres are among the drivers. [1]

The change is not confined to advanced economies. Emerging economies are expected to account for nearly 80% of additional electricity consumption through 2030. China remains the largest single contributor, while India and Southeast Asia are expected to become increasingly important sources of growth. [1]

This matters because electricity is no longer only a utility input. It is a foundational requirement for advanced manufacturing, logistics, communications, computing, healthcare and increasingly electrified services.

Digital growth has a physical footprint.

The IEA estimates that data-centre electricity consumption could rise from approximately 485 TWh in 2025 to around 950 TWh in 2030. AI-focused data-centre consumption is projected to grow faster still. These projections are subject to uncertainty: grid connections, equipment supply, financing conditions and the economics of AI deployment can constrain or alter the pace of expansion. [3]

The implication is not that every digital investment will succeed. It is that digital expansion increasingly depends on power availability, grid capacity, cooling, land, equipment and reliable operations.

Energy security remains a capital-allocation issue.

The IEA estimates global energy investment will reach approximately US$3.4 trillion in 2026. Around US$2.2 trillion is expected to go to renewables, nuclear, grids, storage, low-emissions fuels, efficiency and electrification, while about US$1.2 trillion is expected to go to oil, natural gas and coal. These are 2026 estimates, not final audited outturns. [2]

The investment mix reflects a practical reality: energy systems must balance affordability, reliability, security and emissions objectives. Different regions will pursue different combinations of generation, storage, networks, imports and demand management.

02 / THE STRUCTURAL CHALLENGE

Building assets is only part of the problem.

Infrastructure creates economic value when it delivers reliable services and supports productive activity. Capital expenditure alone does not establish that an asset is well located, efficiently used, properly maintained or generating the expected economic return.

The World Bank’s 2026 infrastructure analysis emphasises that infrastructure gaps remain large and uneven, and that investment quality matters alongside investment volume. It highlights the importance of transparent information on existing assets, costs and returns, and notes that coordinated investment across energy, transport and digital infrastructure can outperform isolated sector approaches. [4]

This points to four connected challenges:

• **Visibility:** Decision-makers need a dependable picture of what physical capacity exists, where it is located and what condition it is in.

• **Performance:** Installed capacity is not the same as available capacity, actual output or continuous service. Performance must be measured over time.

• **Attribution:** Evidence must connect measured activity to the relevant asset, operator, contract, project and economic claim.

• **Financeability:** Lenders and investors need evidence that is sufficiently reliable, comparable and auditable to support underwriting, monitoring and capital allocation.

A measurement is not automatically verification.

A meter, sensor or operational system can produce readings. Those readings still need to be attributed to the correct asset and period, checked for integrity, reconciled against relevant records and interpreted within the limits of the measurement method. The reliability of the evidence chain matters as much as the existence of data.

The objective is not to create another layer of unverified claims. It is to improve the quality of evidence available to people and institutions making consequential decisions.

03 / CAPITAL, TRUST & RESILIENCE

Capital needs confidence in the physical economy.

Energy and infrastructure projects often involve long development periods, multiple counterparties, complex operating conditions and exposure to policy, commodity, construction and technology risks. The quality of project information can affect how risks are assessed, priced, monitored and allocated.

The Asian Development Bank's 2026 publication on private-finance mobilisation reports that multilateral development banks and development finance institutions mobilised a record US$278.5 billion of private finance in 2024. It also reports a widening annual SDG financing gap of US$4 trillion. The figures cover a broad development-finance context; they should not be interpreted as an energy-only funding gap. [5]

The challenge is therefore not simply to attract more capital. It is to improve the conditions under which capital can be deployed productively, especially where project risks, information gaps or fragmented institutional arrangements inhibit investment.

Financial resilience is not the same as predicting a banking crisis.

The IMF's October 2026 Global Financial Stability Report finds that global financial markets have remained resilient, while financial stability risks remain elevated. It identifies potential stress channels including an AI-related asset-price correction, sovereign bond-yield shocks and increasingly sophisticated cyber threats. It also notes that pockets of weaker banks persist, while describing the global banking system overall as broadly resilient. [6]

These findings support a measured conclusion: financial institutions operate in an environment of interconnected risks and need strong controls, transparency and operational resilience. They do not establish that banks are about to fail, nor that physical energy assets can replace money, deposits or regulated financial services.

The role of infrastructure is complementary.

Reliable physical systems support continuity of economic activity. Strong financial institutions mobilise and intermediate capital. Independent evidence and accountable processes can help connect the two. Each has a distinct role; none should be presented as a substitute for the others.

04 / THE INSTITUTIONAL OPPORTUNITY

A more connected approach to infrastructure finance.

The opportunity is to improve the interface between physical productive assets and the institutions that finance, operate, insure, regulate or depend on them.

This requires cooperation across several domains:

Public institutions	Policy alignment, public-interest safeguards, infrastructure priorities and regulatory oversight.
Financial institutions	Capital formation, lending, underwriting, payments and regulated financial services.
Asset owners and operators	Physical delivery, operations, maintenance, performance and contractual accountability.
Technology and engineering providers	Metering, sensors, control systems, cybersecurity, data infrastructure and technical integration.
Investors and development finance	Risk capital, long-term investment, project structuring and mobilisation of private finance.

No single participant needs to own every layer. In fact, clear separation of responsibilities can strengthen the system: operators remain accountable for operating assets; engineers remain responsible for technical systems; regulated institutions retain their regulated roles; public authorities retain statutory powers.

A credible infrastructure-finance framework should therefore prioritise:

- Independent and appropriately scoped evidence.
- Clear attribution of assets, outputs, counterparties and contractual obligations.
- Transparent governance, permissions and accountability.
- Interoperability with existing technical and financial systems.
- Auditability, cybersecurity and clear responsibility for failures or disputes.

05 / THE RHEO PROPOSITION

Making productive capacity measurable for capital.

Rheo's proposed role is to develop a financial infrastructure layer connecting energy, resources and capital. Its focus is the physical economy: the productive assets and infrastructure on which economic activity depends.

The operating logic is expressed in four stages:

01	**MEASURE** Establish relevant measures of physical capacity, output, consumption and performance over defined periods.
02	**VERIFY** Assess the integrity, attribution and consistency of evidence using appropriate independent controls and technical sources.
03	**ACCOUNT** Create traceable records connecting assets, activity, counterparties, contractual terms and performance claims.
04	**INVEST** Make better-quality evidence available to authorised institutions and capital providers for their own assessment and decisions.

What Rheo is — and is not

Rheo is: a proposed independent layer for evidence, accounting and capital governance around physical productive capacity, designed to work with asset owners, operators, technical specialists, financial institutions and public-sector stakeholders.

Rheo is not: an electricity utility, engineering contractor, bank, regulator, cloud provider or replacement for existing financial-market infrastructure. It should not claim to guarantee asset performance, eliminate project risk or confer regulatory approval.

Its credibility will depend on implementation: a clearly defined use case, reliable evidence sources, independent assurance where appropriate, legal and regulatory clarity, cybersecurity, accountable governance and measurable results.

The strategic objective is complementarity, not displacement. Rheo should integrate with the systems and institutions that already perform essential roles, while addressing the gaps between physical performance evidence and capital decisions.

06 / FROM THESIS TO PROOF

The proposition must be demonstrated, not assumed.

A credible pathway begins with a bounded pilot in which the physical asset, measurement method, verification process, data ownership, contractual relationships and intended financial use are all explicit.

A practical sequence

- **Define the use case.** Select a specific infrastructure or energy-finance problem with a real asset owner, operator and prospective institutional user.
- **Establish the evidence chain.** Specify what is measured, by whom, using which equipment and records, over what period, and how readings are attributed and checked.
- **Define governance and liability.** Establish who can submit, verify, approve, amend and rely on records; define dispute handling, access rights and audit trails.
- **Map the regulatory perimeter.** Identify whether any proposed activity touches regulated payments, custody, lending, investment, financial advice, market operation or other regulated services. Use licensed partners where required.
- **Test the financial relevance.** Determine whether the evidence changes underwriting, monitoring, reporting, insurance, procurement or investment decisions in a measurable way.
- **Measure outcomes.** Evaluate accuracy, cost, time, reliability, auditability and whether the pilot improves a real decision. Publish limitations as well as results.

Evidence of progress should include

- A clearly identified asset and accountable operator.
- A documented measurement and verification methodology.
- Reconciled records with traceable sources and timestamps.
- Independent review or assurance appropriate to the use case.
- A defined institutional user and a decision the evidence is intended to support.
- A documented assessment of security, legal, regulatory and operational risks.

This approach avoids the common mistake of treating a broad platform vision as proof of a validated market need. The first proof should be narrow, rigorous and useful to a real decision-maker.

07 / GLOBAL APPLICATION

One structural challenge, different national conditions.

The need to connect productive capacity with capital is global, but the implementation cannot be identical everywhere. Energy mix, grid structure, market design, legal systems, public institutions, currency arrangements and levels of infrastructure maturity differ materially between jurisdictions.

The proposition may be relevant in several settings:

- **Energy-importing economies:** resilience, fuel-price exposure, diversification, storage and demand management.
- **Industrial economies:** reliable power for manufacturing, logistics, advanced materials and high-value production.
- **Digital infrastructure hubs:** power availability, grid connection, cooling, energy efficiency and long-term infrastructure finance.
- **Emerging markets:** asset visibility, project preparation, reliable operating evidence and mobilisation of private capital.
- **Resource-rich economies:** connecting energy and natural resources to domestic productive activity and investment.

Singapore as a strategic starting environment

Singapore is relevant as a potential launch and validation environment because it combines a sophisticated financial centre, established public institutions, significant energy and infrastructure planning, and a regional role in capital and trade. It should be treated as one strategic starting point — not as the sole market or the limit of the thesis.

A Singapore pilot would still need to demonstrate a specific national or commercial use case, fit within applicable regulatory requirements and establish measurable value for participating institutions. Government engagement, commercial partnerships and regulatory dialogue serve different purposes and should not be conflated.

Global ambition requires local legitimacy.

Rheo’s model should be adaptable to local institutions and regulations while maintaining consistent standards for evidence integrity, accountability and capital governance.

08 / CONCLUSION

The next infrastructure question is not only what gets built — but what can be trusted and financed.

The global economy is becoming more dependent on electricity, interconnected infrastructure and capital-intensive physical systems. Demand is rising in many regions, while energy security, grid constraints, financing conditions and operational resilience remain significant considerations.

The strategic opportunity is to improve the quality of the connection between physical productive capacity and financial decision-making. That requires more than technology. It requires credible evidence, clear accountability, institutional cooperation and disciplined implementation.

Important qualifications

This briefing is a strategic perspective, not a forecast of financial-market outcomes, an investment recommendation, a regulatory opinion or a claim that any particular market requires Rheo. Forecasts and estimates are attributed to their original publishers and may change. Rheo's proposed role is a strategic proposition that requires validation through pilots, institutional engagement and independent evidence.

SOURCES & NOTES

Selected primary and institutional sources

[1] International Energy Agency (IEA), *Electricity 2026 — Executive Summary*, 6 February 2026. Global electricity demand forecast, regional drivers and grid/flexibility context.

https://www.iea.org/reports/electricity-2026/executive-summary

[2] International Energy Agency (IEA), *World Energy Investment 2026*, 28 May 2026. Global energy investment estimate and sector allocation. Figures for 2026 are estimates.

https://www.iea.org/reports/world-energy-investment-2026

[3] International Energy Agency (IEA), *Key Questions on Energy and AI — Executive Summary*, 2026. Data-centre electricity demand outlook and uncertainty factors.

https://www.iea.org/reports/key-questions-on-energy-and-ai/executive-summary

[4] World Bank, *Infrastructure Foundations: From Current Assets to Future Growth*, 2026. Infrastructure gaps, investment efficiency, asset data and multisector coordination.

https://www.worldbank.org/en/topic/infrastructure/publication/infrastructure-foundations-from-current-assets-to-future-growth

[5] Asian Development Bank and partner multilateral development banks, *Mobilization of Private Finance by Multilateral Development Banks and Development Finance Institutions 2024 Joint Report*, published May 2026. Private-finance mobilisation and SDG financing-gap context.

https://www.adb.org/documents/mobilization-private-finance-mdbs-dfis-2024

[6] International Monetary Fund (IMF), *Global Financial Stability Report: Resilience, Risk, and Transformation*, October 2026. Financial stability risks and resilience assessment.

https://www.imf.org/en/Publications/GFSR/Issues/2026/10/13/global-financial-stability-report-october-2026

Editorial note

This document deliberately distinguishes reported facts and institutional forecasts from strategic interpretation. Forecasts are labelled as such. It does not infer a banking crisis from market volatility, nor imply that energy assets replace monetary or banking systems. It positions energy and physical infrastructure as foundations of economic activity, with financial institutions retaining their established and regulated functions.

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